

Moment of a Force (Torque)

Moment about a point · perpendicular & inclined forces · sign convention

Physics — Grade 11
Revision Sheet
g is not needed here

1 The turning effect of a force

When you push a door, the door does not move forward — it **rotates** about its hinge. A force can therefore do two different things: it can **move** a body (translation), or it can **turn** it about a fixed point (rotation).

Try the same push in three different places on the door (Fig. 1). The force is exactly the same, but the turning effect is completely different. Push very close to the hinge and the door hardly moves; push at the far edge and it swings easily.

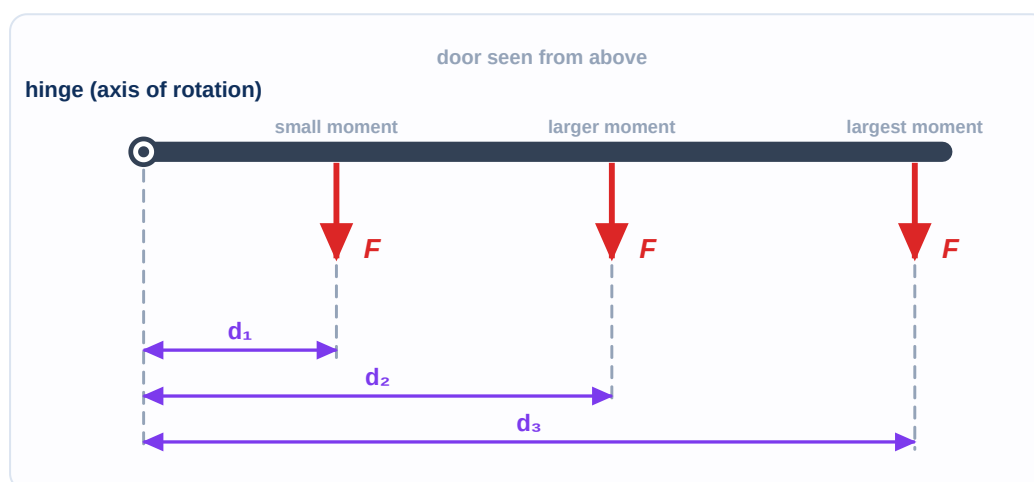


Fig. 1 — The same force F produces a different turning effect when its distance from the hinge changes.

Conclusion

The turning effect of a force depends on **two** things: the **magnitude of the force** (F) and the **perpendicular distance** between the force and the axis of rotation (d). This combined quantity is called the **moment of the force**.

التأثير الدوراني يعتمد على مقدار القوة وعلى المسافة العمودية من محور الدوران.

2 Definition of the moment of a force

Definition: The moment of a force about a point is *the product of the force and the perpendicular distance between the point and the line of action of the force*.

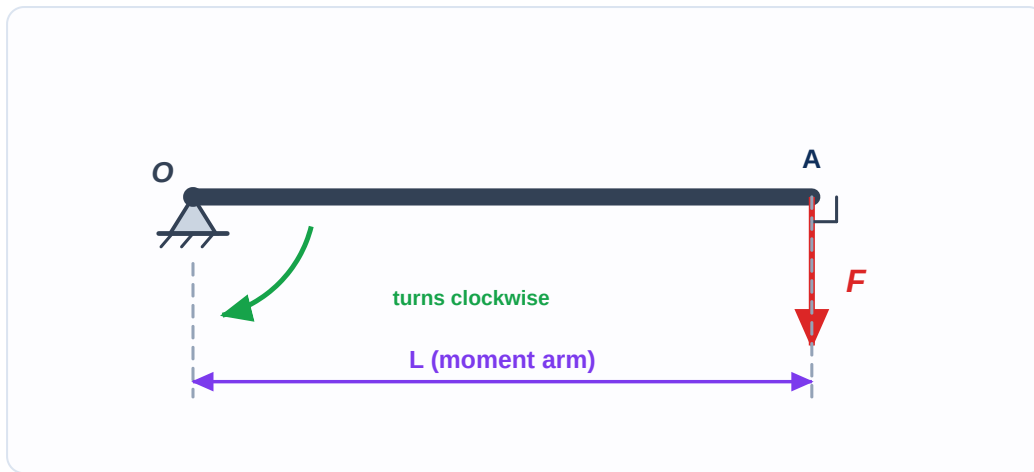


Fig. 2 — Moment of the force F about the point O . $M = F \times L$

$$M = F \times L$$

M = moment ($\text{N}\cdot\text{m}$) | F = force (N) | L = perpendicular distance from the point to the line of action (m)

Quantity	Symbol	SI unit	Type
Force	F	newton (N)	vector
Moment arm	L or d	metre (m)	scalar (a distance)
Moment of a force	M	newton · metre ($\text{N}\cdot\text{m}$)	vector

Careful

The unit of the moment is **$\text{N}\cdot\text{m}$** , and it is **never** written as a joule, although $1 \text{ N} \times 1 \text{ m}$ also appears in the unit of work. Moment and work are completely different physical quantities.

3 The line of action and the moment arm

The **line of action** of a force is the straight line drawn through the force in the direction of the force, extended as far as you like in both directions. The **moment arm** d is the **shortest** (that is, the perpendicular) distance from the point O to this line.

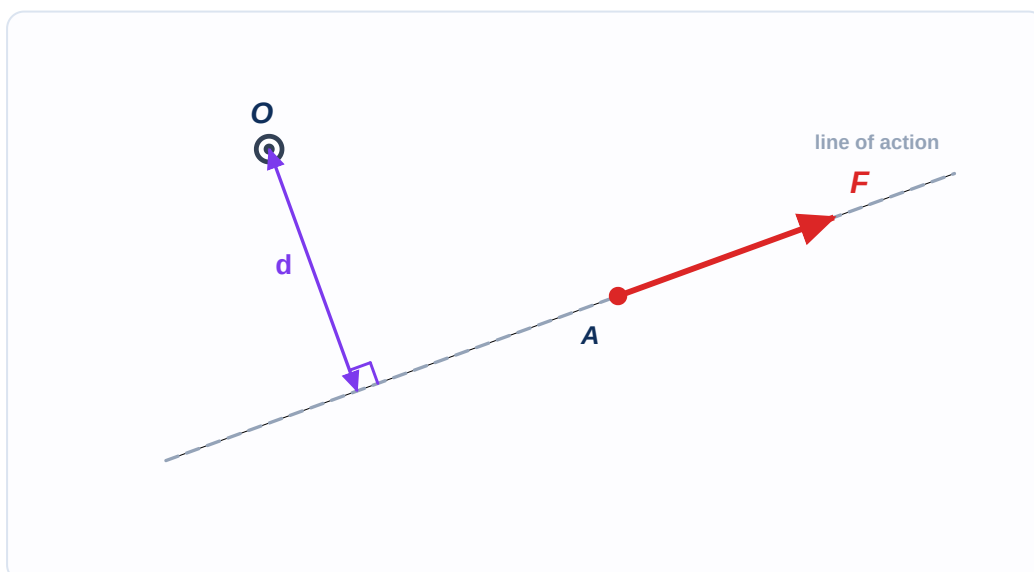


Fig. 3 — The moment arm d is the **PERPENDICULAR** distance from O to the line of action of F .

How to find the moment arm

1. Extend the line of action of the force (dotted line).
2. From the point O drop a perpendicular onto that line.
3. The length of this perpendicular is d .

الذراع هو دائماً العمود المرسوم من النقطة على خط عمل القوة — مش المسافة المائلة.

4 Sign convention — the direction of the moment

The moment is a **vector** quantity, so it must be given a direction. In problems about one point we do not need arrows; we only need to say whether the force turns the body **anticlockwise** or **clockwise**:

anticlockwise → **M is positive (+)**

clockwise → **M is negative (-)**

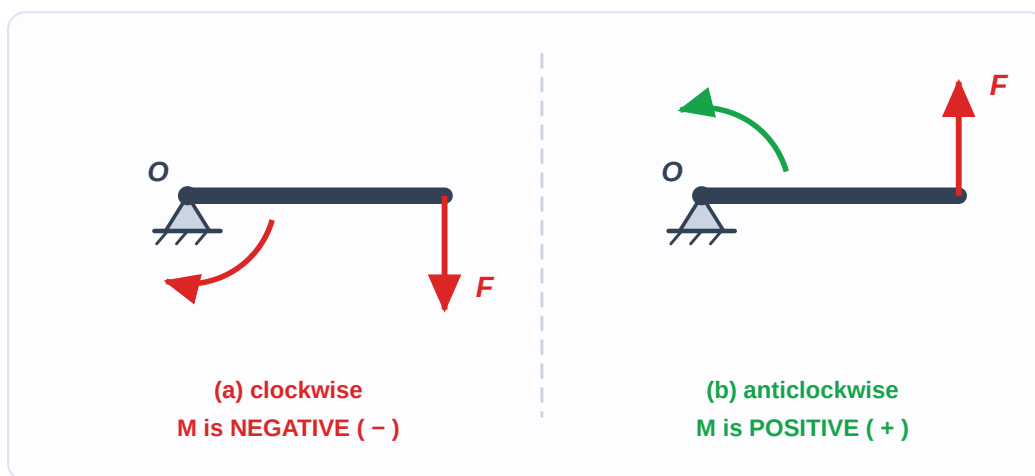


Fig. 4 — Sign convention used everywhere in this sheet.

Imagine holding the body at O and letting the force act alone: look at the direction in which the body starts to turn, then give the moment its sign.

5 When is the moment equal to zero?

Since $M = F \times L$, the moment vanishes in two cases:

- $F = 0$ — there is no force at all.
- $L = 0$ — the **line of action of the force passes through the point O**. This happens when the force is directed along the rod, or is applied at the axis itself.

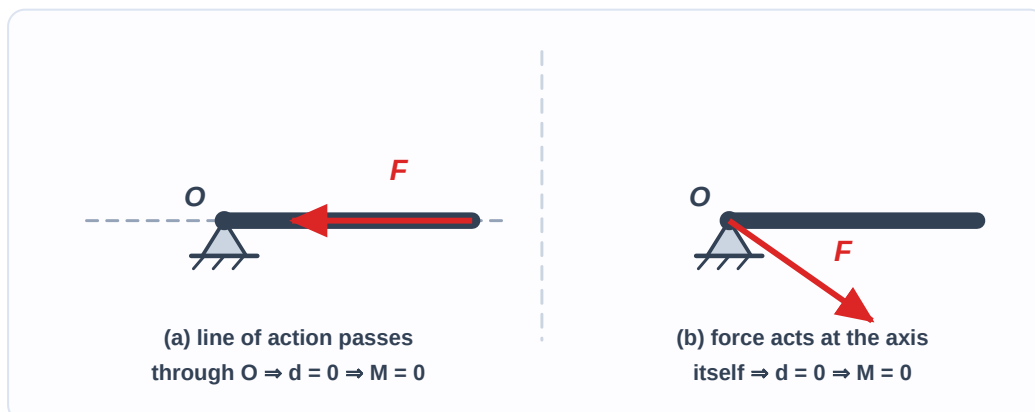


Fig. 5 — Two cases in which the moment about O is zero.

Everyday check

Pushing a door exactly at its hinge, or pushing it along its own plane (edge-on), never opens it — the line of action passes through the hinge, so $M = 0$.

6 Case 1: the force is perpendicular to the rod

This is the simplest — and the most common — case. If the force is perpendicular ($\perp$) to the rod, then the length of the rod **is itself** the moment arm.

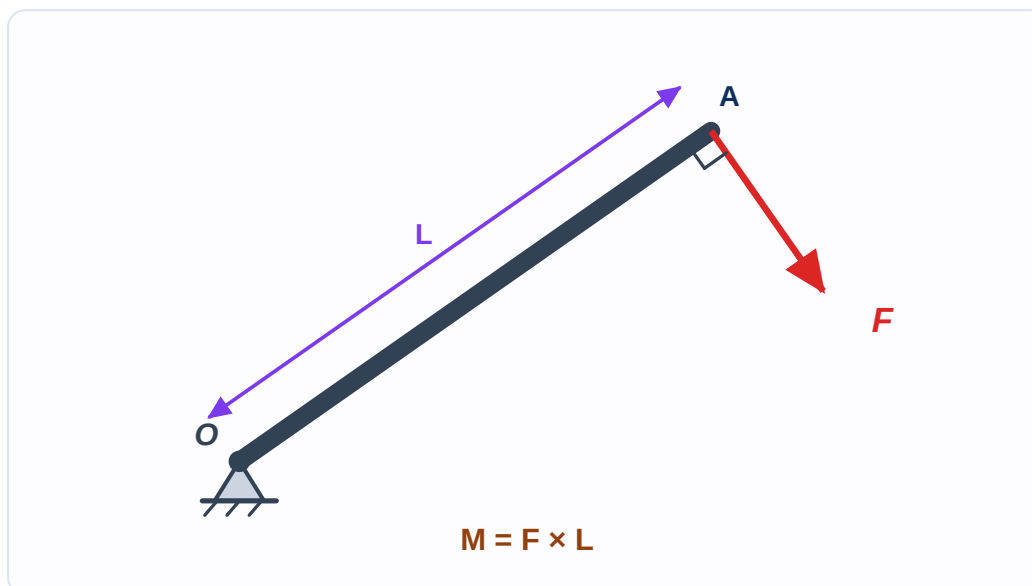


Fig. 6 — CASE 1: the force is perpendicular to the rod, so the moment arm is the length OA itself.

$$M = F \times L \quad (\theta = 90^\circ)$$

this gives the **largest possible moment** for a given force and a given length

Note

“Perpendicular” means perpendicular **to the rod**, not necessarily vertical. In Fig. 6 the rod is tilted, yet the force is still $\perp$ to it, so $M = F L$.

7 Case 2: the force is inclined at an angle θ to the rod

Method 1 — resolve the force

Resolve F into two components: one **perpendicular to the rod** ($F \sin\theta$) and one **along the rod** ($F \cos\theta$).

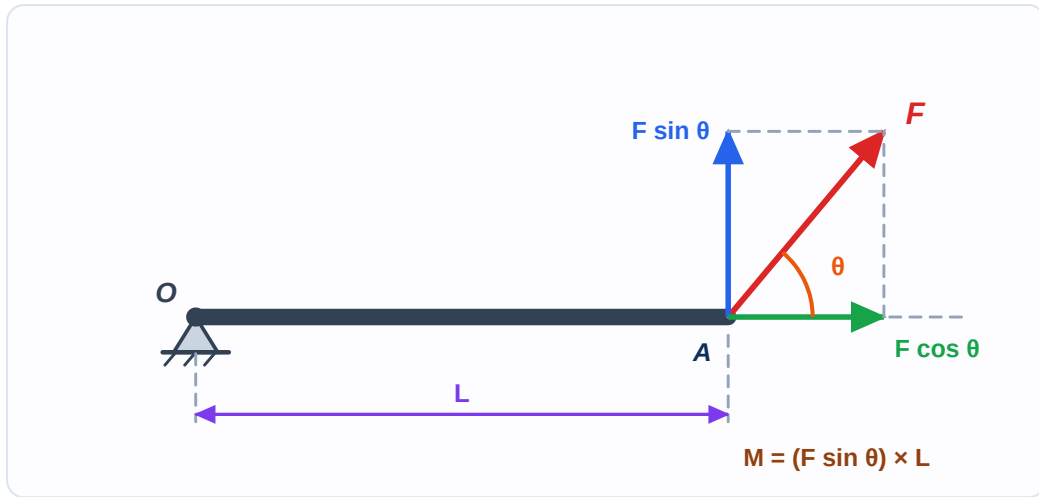


Fig. 7 — CASE 2 (method 1): resolve F into a component $\perp$ to the rod and a component along the rod.

- The component $F \sin \theta$ is perpendicular to the rod $\rightarrow$ it **turns** the rod. Its moment is $(F \sin \theta) \times L$.
- The component $F \cos \theta$ lies along the rod, so its line of action passes through O $\rightarrow$ its moment is **zero**. It only pulls or pushes the rod.

Method 2 — use the perpendicular distance

Instead of resolving the force, keep the force as it is and find the perpendicular distance from O to its line of action. From the right-angled triangle, that distance is $d = L \sin \theta$.

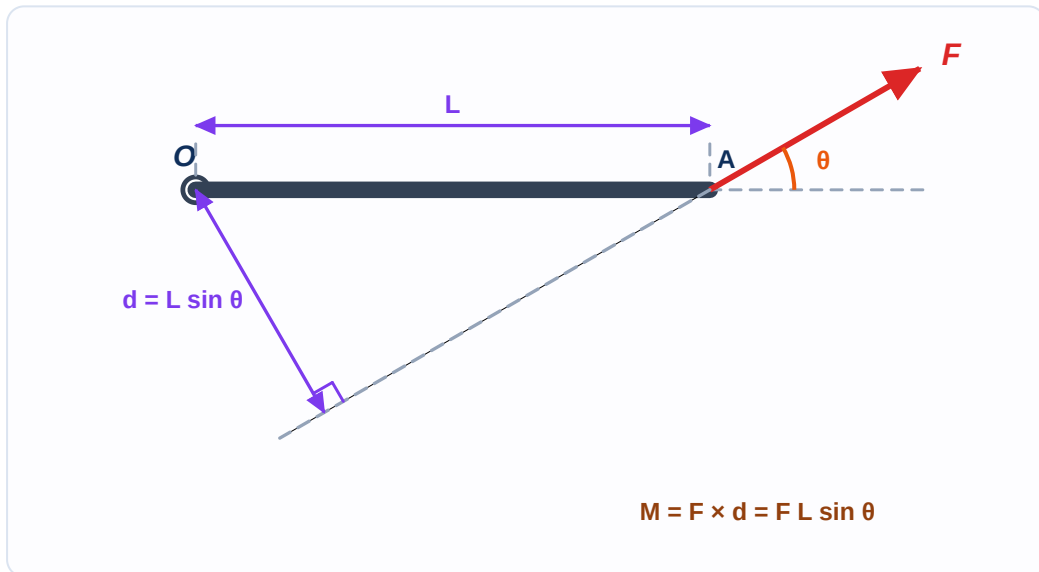


Fig. 8 — CASE 2 (method 2): drop a perpendicular from O onto the line of action: $d = L \sin \theta$.

$$M = F \times L \times \sin \theta$$

θ = the angle between the force and the rod · both methods give exactly the same result

القانون واحد: $M = F L \sin \theta$ — ولو القوة عمودية ($\theta = 90^\circ$) تبقى $\sin \theta = 1$ ويرجع القانون $M = F L$.

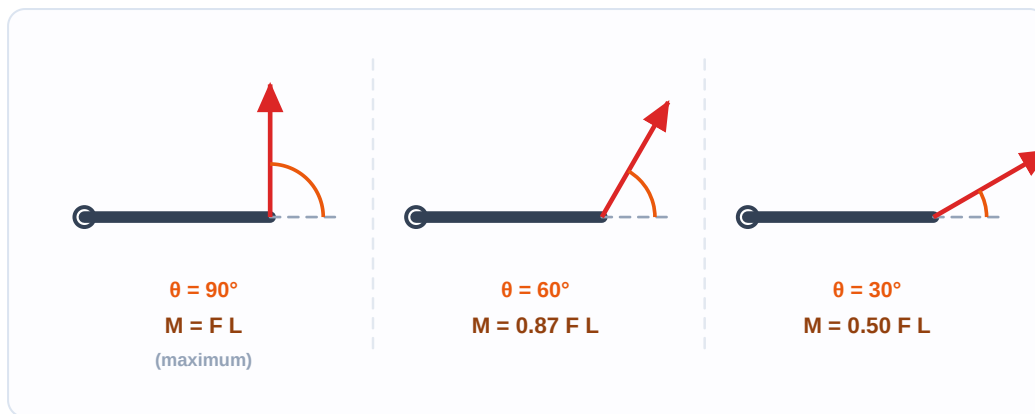


Fig. 9 — Same force F , same length L , different angle θ : the moment changes with $\sin \theta$.

θ	0°	30°	45°	60°	90°	150°	180°
$\sin \theta$	0	0.5	0.707	0.866	1	0.5	0
M	0	$0.5 FL$	$0.71 FL$	$0.87 FL$	$FL \text{ (max)}$	$0.5 FL$	0

8 Several forces acting on the same body

When more than one force acts on a body, the total (resultant) moment about a point is the **algebraic sum** of the separate moments — each one taken with its own sign.

$$M_{\text{net}} = \Sigma M = M_1 + M_2 + M_3 + \dots$$

put a + sign for every anticlockwise moment and a – sign for every clockwise moment

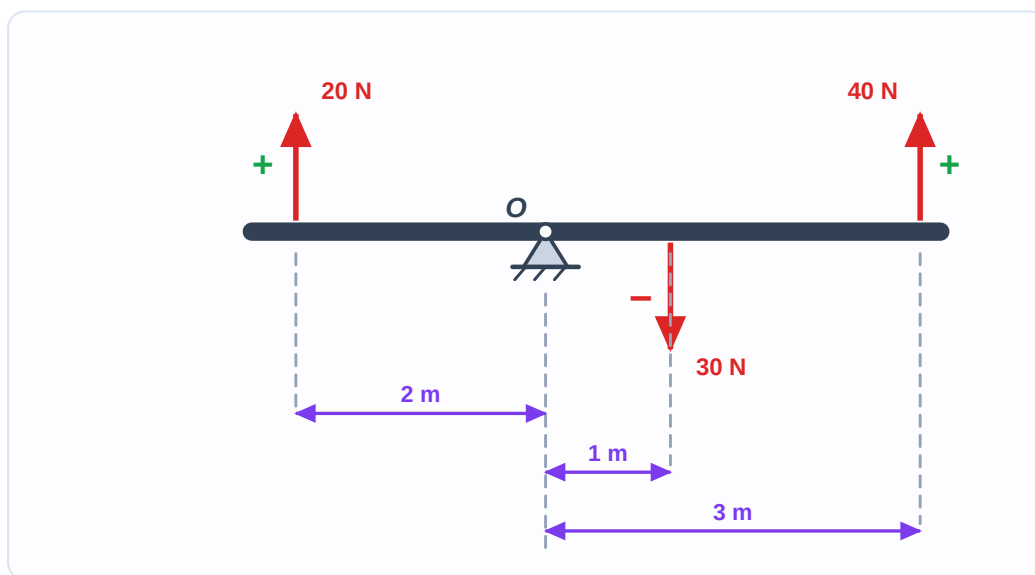


Fig. 10 — Several forces acting on the same bar: add their moments about O with the correct sign.

$$\begin{aligned}
 M_{\text{net}} &= (+20 \times 2) + (-30 \times 1) + (+40 \times 3) \\
 &= +40 - 30 + 120 = \mathbf{+130 \text{ N}\cdot\text{m}} \quad (\text{the } + \text{ sign means: anticlockwise})
 \end{aligned}$$

Reading the answer

The **size** of the number tells you how strong the turning effect is; the **sign** tells you the direction of the turn. Always finish your answer with the word *clockwise* or *anticlockwise*.

★ Key facts you must memorise

- $M = F \times L$ (force $\perp$ to the rod) | $M = F \times L \sin\theta$ (force inclined at θ)
- Unit: **N·m**. Moment is a **vector** quantity.
- **Anticlockwise** = + , **clockwise** = -
- $M = 0$ when the line of action of the force passes through the point.
- M is a **maximum** when $\theta = 90^\circ$ and **zero** when $\theta = 0^\circ$ or 180° .
- Total moment about a point = algebraic sum of the moments (with signs).

Common mistakes

- Using the slanted distance instead of the **perpendicular** distance (forgetting $\sin\theta$).
- Using $\cos\theta$ instead of $\sin\theta$. Remember: the component that **turns** the rod is the perpendicular one.
- Leaving the length in centimetres: 20 cm = **0.2 m**, not 20.
- Adding moments without their signs.
- Giving the answer as a number only, with no direction.

حوّل السنتيمتر لمتر، وخذ بالك من الإشارة، واكتب الاتجاه في الإجابة النهائية.

9 Solved examples

Example 1 — a perpendicular force (spanner)

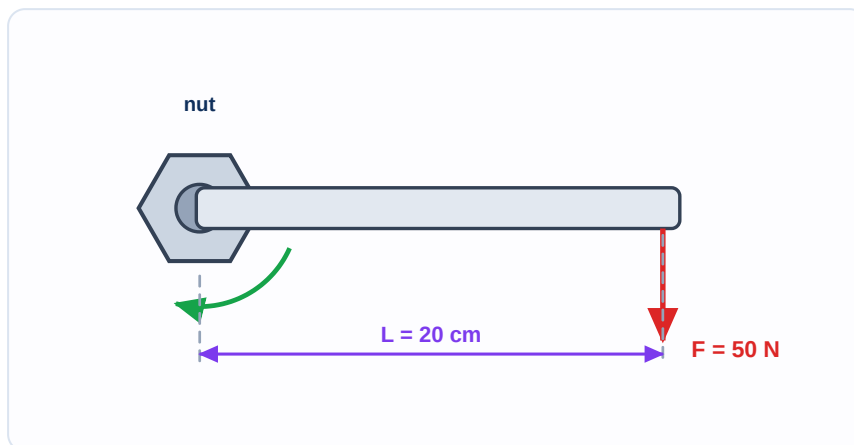


Fig. 11 — the force is perpendicular to the handle of the spanner.

Given: $F = 50 \text{ N}$ perpendicular to the spanner, $L = 20 \text{ cm}$. Find the moment about the centre of the nut.

Solution

The force is $\perp$ to the handle, so the moment arm is the length itself. First change cm into m:

$$L = 20 \text{ cm} = 0.20 \text{ m}$$

$$M = F \times L = 50 \times 0.20 = 10 \text{ N}\cdot\text{m}$$

The spanner turns clockwise $\rightarrow$ the sign is negative.

$$M = -10 \text{ N}\cdot\text{m} \text{ (10 N}\cdot\text{m clockwise)}$$

Example 2 — an inclined force on a rod

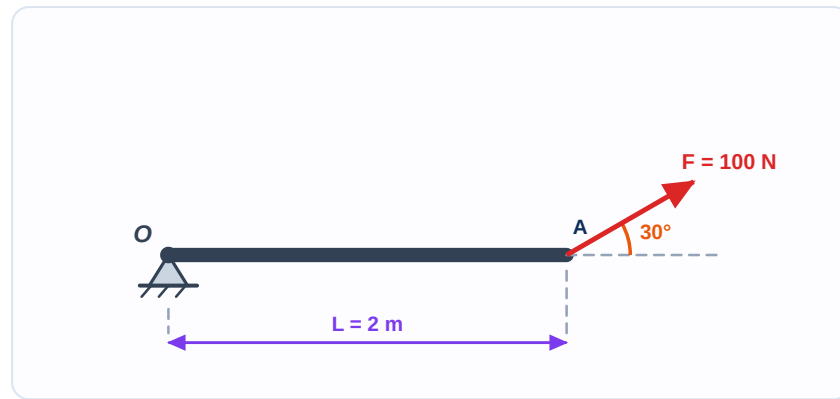


Fig. 12 — the force makes 30° with the rod.

Given: A rod OA of length 2 m is hinged at O. A force of 100 N acts at A at 30° to the rod.

Solution

$$M = F \times L \times \sin \theta$$

$$M = 100 \times 2 \times \sin 30^\circ$$

$$M = 100 \times 2 \times 0.5 = 100\text{ N}\cdot\text{m}$$

The rod turns anticlockwise $\rightarrow$ positive.

Check with method 1: $F \sin \theta = 100 \times 0.5 = 50\text{ N}$, and $50 \times 2 = 100\text{ N}\cdot\text{m}$. Same answer.

$$M = + 100\text{ N}\cdot\text{m} \text{ (anticlockwise)}$$

Example 3 — pushing a door at an angle



Fig. 13 — the push is not perpendicular to the door.

Given: A door of width 0.8 m is pushed at its outer edge with a force of 40 N making 60° with the door.

Solution

$$M = F L \sin \theta = 40 \times 0.8 \times \sin 60^\circ$$

$$M = 40 \times 0.8 \times 0.866 = 27.7 \text{ N}\cdot\text{m}$$

If the same force were applied perpendicular to the door, the moment would be $40 \times 0.8 = 32 \text{ N}\cdot\text{m}$ — that is why we always push a door perpendicular to its surface.

$$M = 27.7 \text{ N}\cdot\text{m} \text{ (clockwise in the figure)}$$

Example 4 — the net moment of three forces

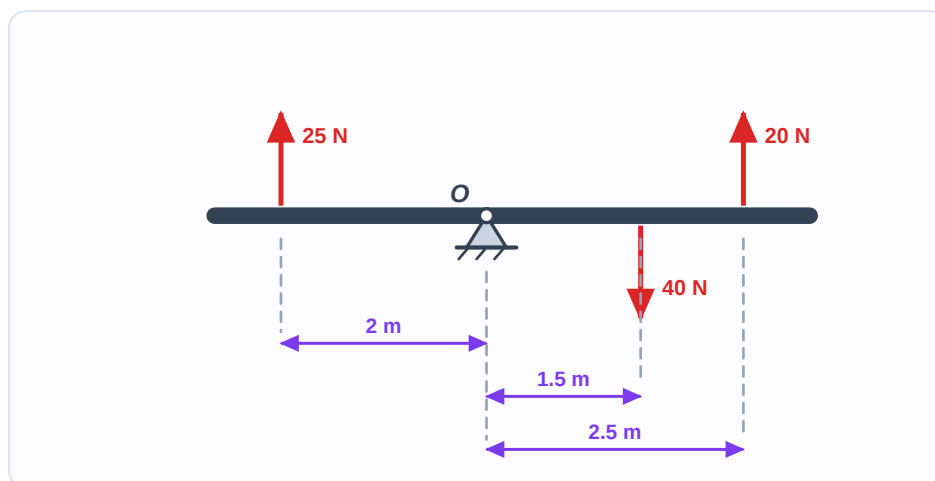


Fig. 14 — three forces act on the same bar.

Given: Find the resultant moment about O for the bar shown in Fig. 14.

Solution

Take each force on its own and decide its sense of rotation about O:

25 N up, 2 m to the LEFT	→ clockwise	→ $M_1 = - 25 \times 2 = - 50 \text{ N}\cdot\text{m}$
40 N down, 1.5 m to the RIGHT	→ clockwise	→ $M_2 = - 40 \times 1.5 = - 60 \text{ N}\cdot\text{m}$
20 N up, 2.5 m to the RIGHT	→ anticlockwise	→ $M_3 = + 20 \times 2.5 = + 50 \text{ N}\cdot\text{m}$

$$M(\text{net}) = -50 - 60 + 50 = - 60 \text{ N}\cdot\text{m}$$

$$M_{\text{net}} = - 60 \text{ N}\cdot\text{m} \text{ (60 N}\cdot\text{m clockwise)}$$

Example 5 — a tangential force on a wheel

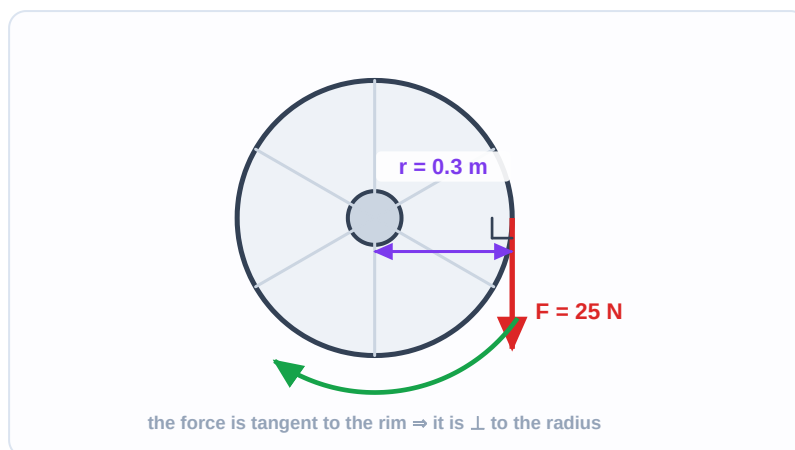


Fig. 15 — a force tangent to the rim is perpendicular to the radius.

Given: A force of 25 N acts at the rim of a wheel of radius 0.3 m, along the tangent.

Solution

A tangent is always perpendicular to the radius at the point of contact, so $\theta = 90^\circ$ and the moment arm is the radius itself.

$$M = F \times r = 25 \times 0.3 = 7.5 \text{ N}\cdot\text{m} \quad (\text{clockwise})$$

$$M = - 7.5 \text{ N}\cdot\text{m} \quad (7.5 \text{ N}\cdot\text{m} \text{ clockwise})$$

Example 6 — finding the force (an inverse problem)

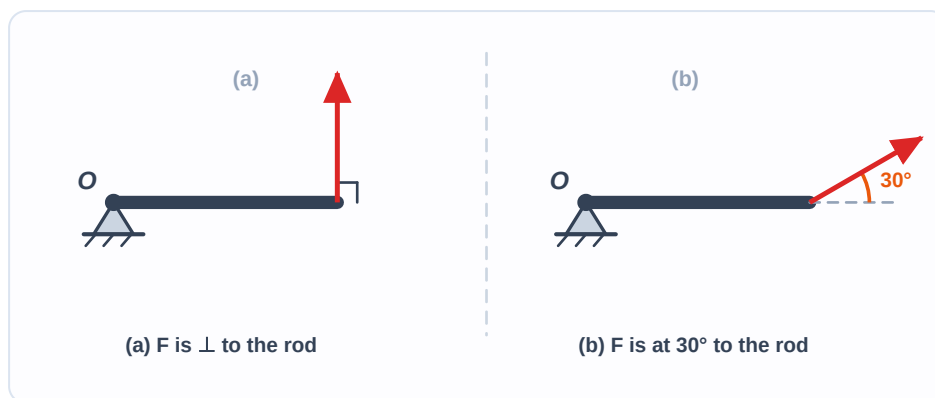


Fig. 16 — the same moment is required in two different situations.

Given: A moment of $48 \text{ N}\cdot\text{m}$ is required about O , and the rod is 1.6 m long. Find the force needed (a) when it is perpendicular to the rod, (b) when it makes 30° with the rod.

Solution

$$(a) \quad M = F L \quad \rightarrow \quad F = M / L = 48 / 1.6 = 30 \text{ N}$$

$$(b) \quad M = F L \sin\theta \quad \rightarrow \quad F = M / (L \sin\theta)$$
$$F = 48 / (1.6 \times \sin 30^\circ) = 48 / (1.6 \times 0.5) = 48 / 0.8 = 60 \text{ N}$$

The inclined force must be **twice** as large to produce the same turning effect.

(a) $F = 30 \text{ N}$ | (b) $F = 60 \text{ N}$

Example 7 — an obtuse angle

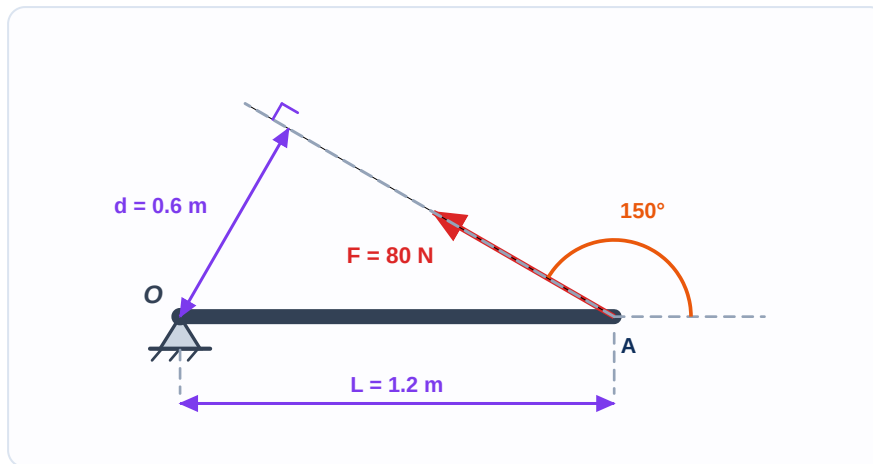


Fig. 17 — An obtuse angle: $\sin 150^\circ = \sin 30^\circ = 0.5$

Given: A force of 80 N acts at the end A of a rod $OA = 1.2\text{ m}$, making an angle of 150° with the rod.

Solution

$$\begin{aligned} M &= F L \sin \theta = 80 \times 1.2 \times \sin 150^\circ \\ \sin 150^\circ &= \sin (180^\circ - 150^\circ) = \sin 30^\circ = 0.5 \\ M &= 80 \times 1.2 \times 0.5 = 48\text{ N}\cdot\text{m} \end{aligned}$$

Check: the perpendicular distance is $d = L \sin 150^\circ = 1.2 \times 0.5 = 0.6\text{ m}$, and $M = F \times d = 80 \times 0.6 = 48\text{ N}\cdot\text{m}$.

$$M = + 48\text{ N}\cdot\text{m} \text{ (anticlockwise)}$$

10 Exercises

18 questions. Give every answer with its **unit**, its **sign** and the **sense of rotation**. Take the anticlockwise sense as positive.

Q1 A force of 30 N acts at the end A of a rod OA, perpendicular to the rod. If $OA = 0.6$ m, find the moment of the force about O.

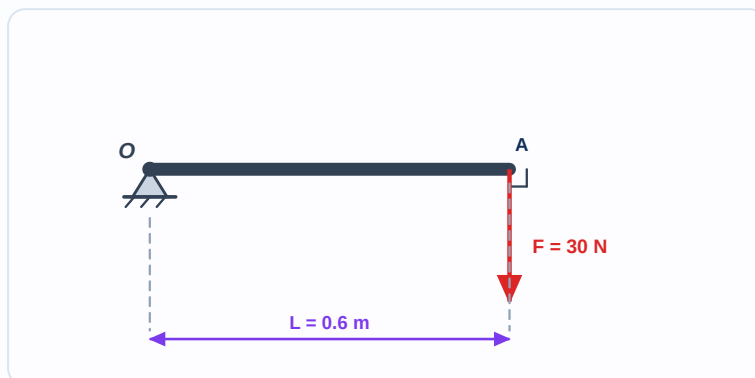


Fig. 18

Q2 A spanner of length 15 cm is used to tighten a nut. What force, applied perpendicular to the spanner, produces a moment of 12 N·m about the nut?

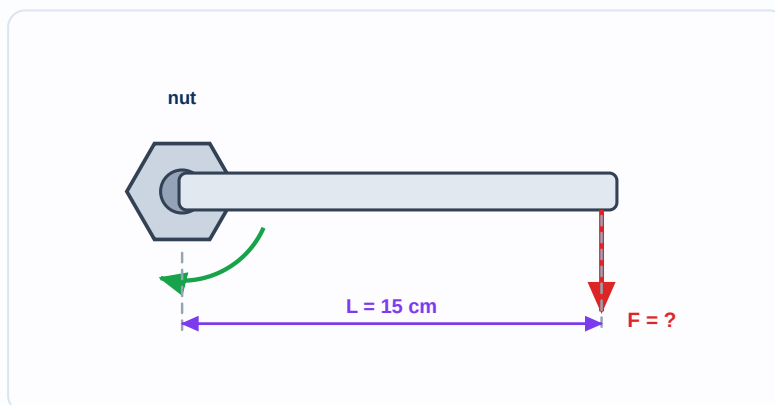


Fig. 19

Q3 A force of 60 N acts at the end of a rod of length 1.4 m, making an angle of 45° with the rod. Find the moment of this force about the hinge O.

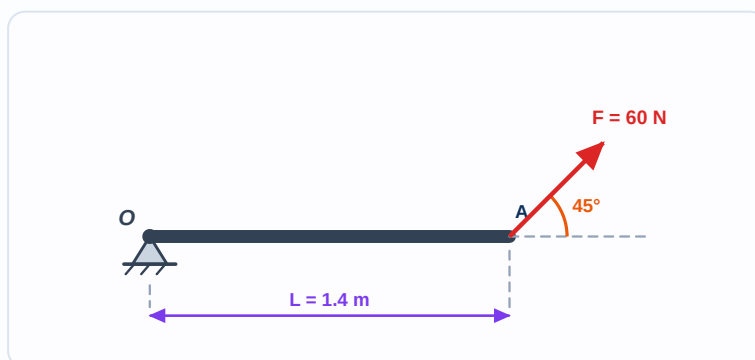


Fig. 20

- Q4** A door of width 0.9 m is pushed at its edge by a force of 25 N inclined at 30° to the door. Find the moment about the hinge, and state how much larger the moment would be if the force were perpendicular.



Fig. 21

- Q5** Find the resultant moment about O of the two forces acting on the bar.

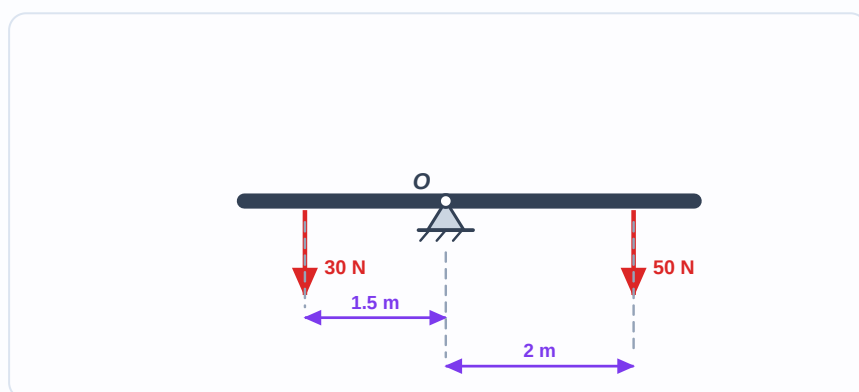


Fig. 22

- Q6** Find the resultant moment about O of the three forces acting on the bar, and state its direction.

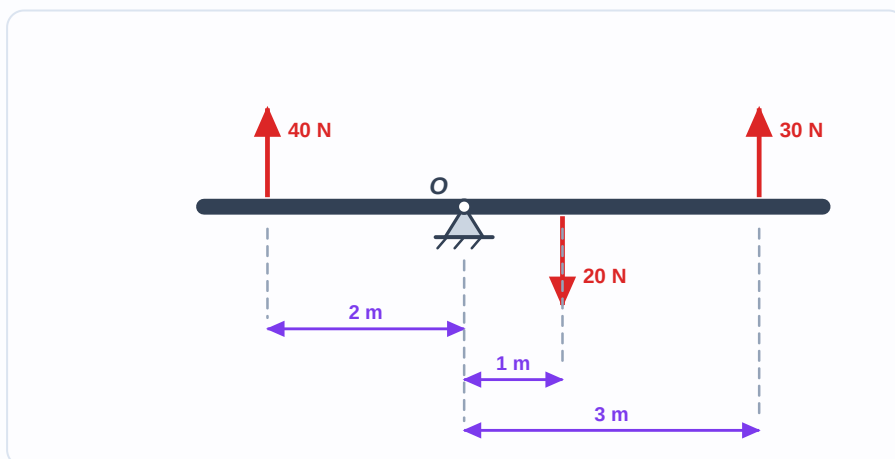


Fig. 23

Q7 A force of 70 N is applied at the end A of the rod OA, directed **along** the rod as shown. Find its moment about O and explain your answer.

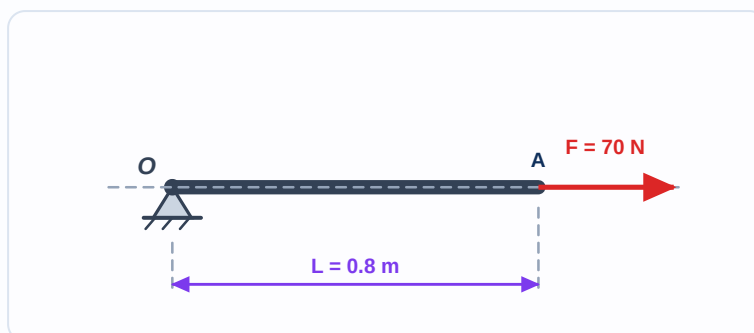


Fig. 24

Q8 A force of 40 N acts along the tangent at the rim of a wheel of radius 25 cm. Find the moment of the force about the centre of the wheel.

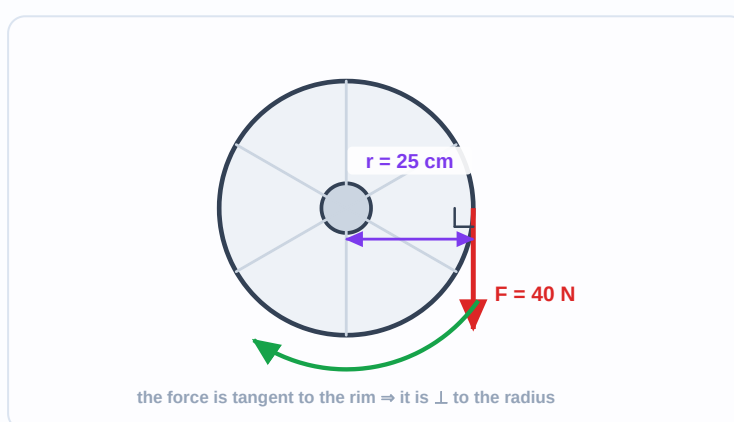


Fig. 25

Q9 The same force of 40 N acts at the end of the same rod ($L = 1 \text{ m}$) in the two positions shown. Find the moment in each case, then find the ratio $M_a : M_b$.

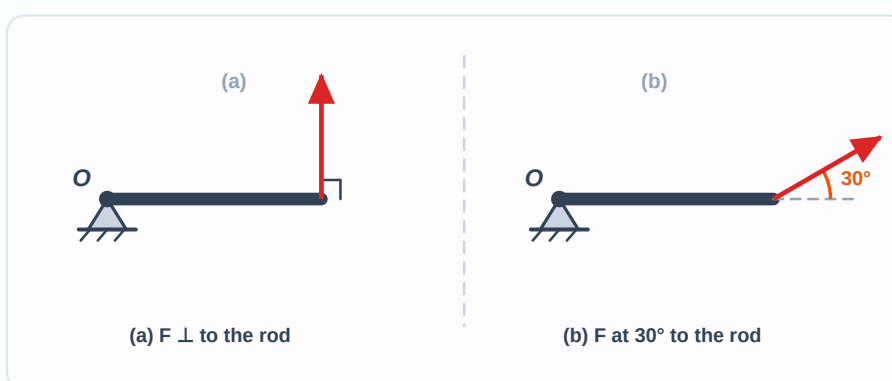


Fig. 26

Q10 A force of 50 N acts at the end of a rod of length 1.2 m and produces a moment of 30 N·m about O. Find the angle θ between the force and the rod.

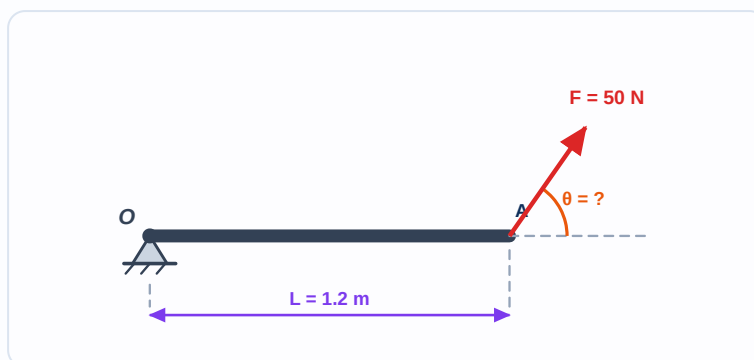


Fig. 27

Q11 A force of 120 N acts at the end A of a rod OA of length 0.5 m, perpendicular to the rod as shown. Find the moment of the force about O.

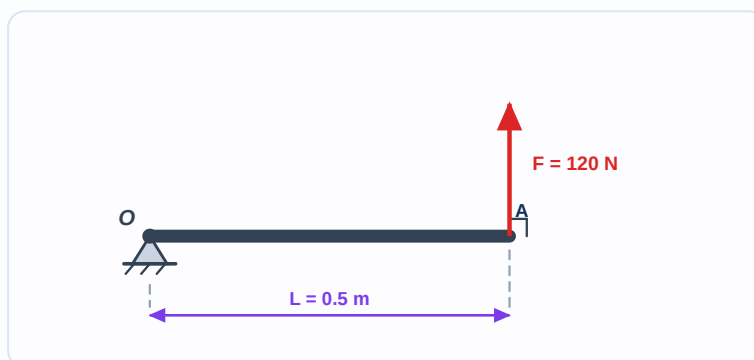


Fig. 28

Q12 Find the resultant moment about O of the two forces acting on the bar, and state its sense of rotation.

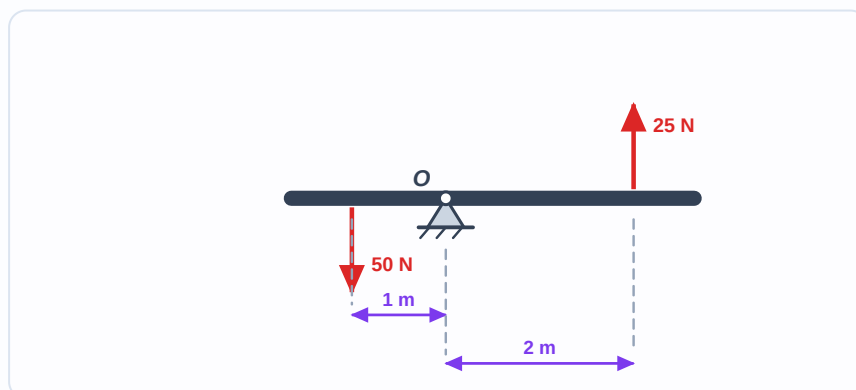


Fig. 29

Q13 A force of 100 N acts at the end of a rod of length 1.5 m, making an angle of 60° with the rod. Find the moment of the force about the hinge O.

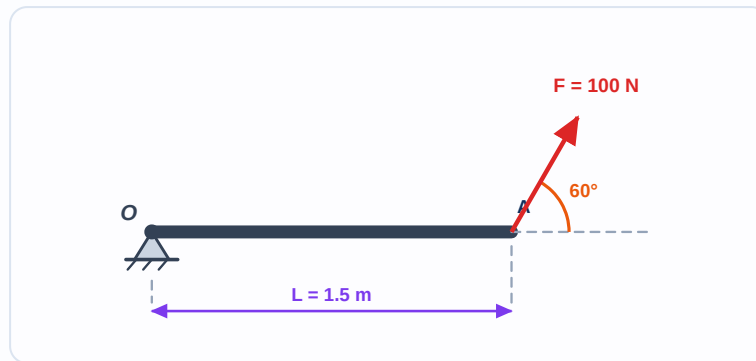


Fig. 30

Q14 A force of 90 N acting perpendicular to a rod produces a moment of $36 \text{ N}\cdot\text{m}$ about the hinge. Find the length of the rod.

Q15 A door can be pushed in two ways: (i) with a force of 50 N perpendicular to the door at a distance of 0.6 m from the hinge, or (ii) with a force of 30 N perpendicular to the door at a distance of 1.0 m from the hinge. Calculate the moment in each case and say which way opens the door more easily.

Q16 A force of 15 N acts along the tangent at the rim of a wheel of radius 0.4 m. Find the moment of the force about the centre of the wheel.

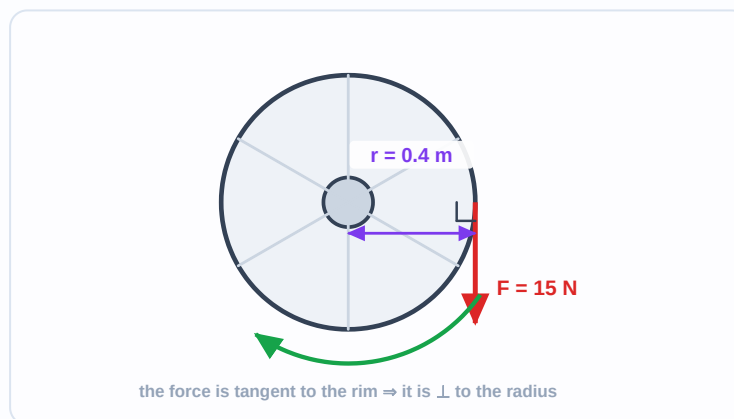


Fig. 31

Q17 A force of 80 N acts at the end of a rod of length 2 m, making an angle of 120° with the rod. Find the moment of the force about O. (Remember: $\sin 120^\circ = \sin 60^\circ$.)

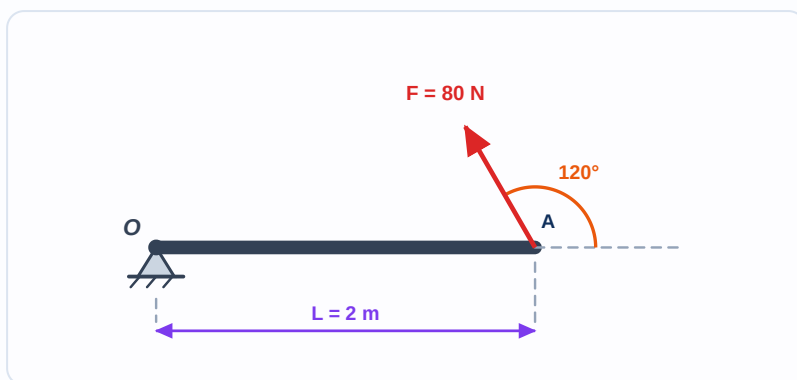


Fig. 32

Q18 Four forces act on the bar shown. Find the resultant moment about O and state whether the bar turns clockwise or anticlockwise.

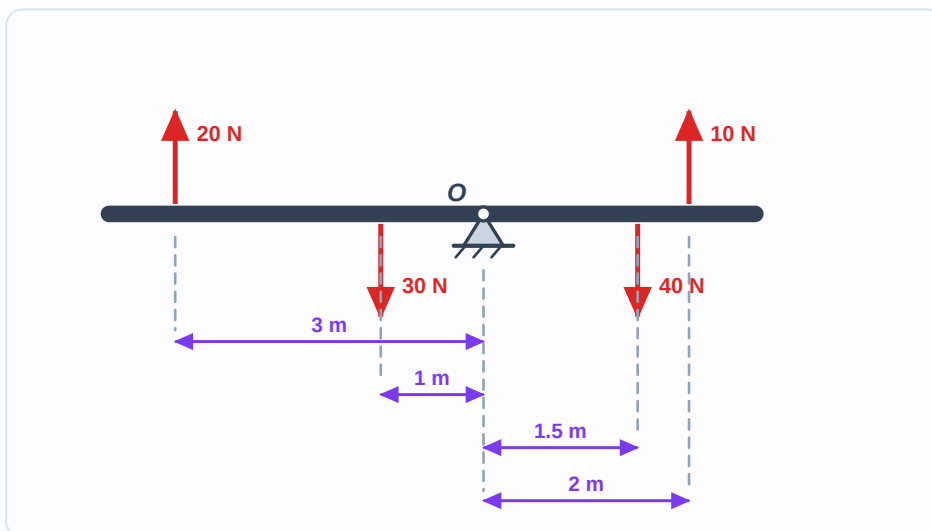


Fig. 33

11 Model answer — final answers only

Use this page to correct the exercises. Only the final answer is given, so the student must write every step of the working by himself.

Q	Final answer	Q	Final answer
1	$-18 \text{ N}\cdot\text{m}$ (clockwise)	2	80 N
3	$+59.4 \text{ N}\cdot\text{m}$ (anticlockwise)	4	$11.25 \text{ N}\cdot\text{m}$; perpendicular force gives $22.5 \text{ N}\cdot\text{m}$ (twice as large)
5	$-55 \text{ N}\cdot\text{m}$ (clockwise)	6	$-10 \text{ N}\cdot\text{m}$ (clockwise)
7	$M = 0$	8	$10 \text{ N}\cdot\text{m}$
9	$M_a = 40 \text{ N}\cdot\text{m}$, $M_b = 20 \text{ N}\cdot\text{m}$, ratio = 2 : 1	10	$\theta = 30^\circ$
11	$+60 \text{ N}\cdot\text{m}$ (anticlockwise)	12	$+100 \text{ N}\cdot\text{m}$ (anticlockwise)
13	$+129.9 \text{ N}\cdot\text{m}$ (anticlockwise)	14	$L = 0.4 \text{ m}$
15	(i) $30 \text{ N}\cdot\text{m}$, (ii) $30 \text{ N}\cdot\text{m}$ — the two ways are equally easy	16	$6 \text{ N}\cdot\text{m}$
17	$+138.6 \text{ N}\cdot\text{m}$ (anticlockwise)	18	$-70 \text{ N}\cdot\text{m}$ (clockwise)